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Architecture Selection – Incident Escalation & On-Call Rotation Engine

1. Architecture Selection

I chose Microservices Architecture for the Incident Escalation & On-Call Rotation Engine.

2. Architectural Choice

The system is designed using a Microservices Architecture in which major functionalities are divided into independent services. The main components are Alert Ingestion Service, Incident & Escalation Service, On-Call Rotation Service, Notification Service, and Post-Mortem Service.

3. Reason 1 – Independent Functionality

The system contains several distinct responsibilities such as incident ingestion, on-call rotation, incident escalation, notification delivery, and post-mortem management. Microservices allow these responsibilities to be implemented and maintained as separate services.

4. Reason 2 – Scalability

Different services may experience different workloads during incidents. Microservices allow heavily used services, such as the Notification Service and Incident & Escalation Service, to be scaled independently without scaling the entire system.

5. Security Advantage

Each service can enforce authentication and authorization at its API boundary. This helps restrict access to incident information, on-call schedules, and management functions to authorized users and services.

6. Performance Benefit

Services involved in alert ingestion and notification can be scaled independently during high incident loads. This helps the system respond quickly to incoming alerts and send escalation notifications within the required response time.

7. Components

Component	Responsibility
Alert Ingestion Service	Receives and records incident alerts
Incident & Escalation Service	Manages incidents and escalation logic
On-Call Rotation Service	Identifies active and secondary on-call engineers
Notification Service	Sends incident and escalation notifications
Post-Mortem Service	Tracks and reviews post-incident information

8. Interfaces

Interface	Purpose
Alert Ingestion → Incident & Escalation	Incident Alert
Incident & Escalation → On-Call Rotation	Get Active On-Call Engineer
Incident & Escalation → Notification	Send Notification
Incident & Escalation → Post-Mortem	Store/Track Post-Mortem